JOB DESCRIPTION

Embedded Firmware Engineer (C/C++)

Company: Voltrix Electronics (fictional)
Location: Bengaluru / On-site
Experience: 2-5 years

About the role

Voltrix designs motor controllers and smart sensors for industrial and automotive customers. We are looking for a Firmware Engineer to write reliable, real-time code that runs directly on our hardware.

Responsibilities

- Develop firmware in C and C++ for ARM Cortex-M microcontrollers such as STM32.
- Write and test peripheral drivers for UART, SPI, I2C and CAN bus.
- Use an RTOS such as FreeRTOS for scheduling, queues and semaphores.
- Debug hardware and software issues using GDB, JTAG, an oscilloscope and a logic analyzer.
- Follow MISRA C coding standards and use static analysis tools.
- Write test scripts in Python and maintain builds with Makefile or CMake.
- Work with hardware engineers on board bring-up.

Required skills

- Embedded C and C++
- ARM Cortex-M microcontrollers
- RTOS concepts
- UART, SPI, I2C and CAN protocols
- Debugging with GDB and JTAG
- Git

Good to have

- Embedded Linux and device drivers
- MISRA C
- Python scripting
- Experience with automotive or industrial products